Extensive-form games

Example

- Consider the following game:

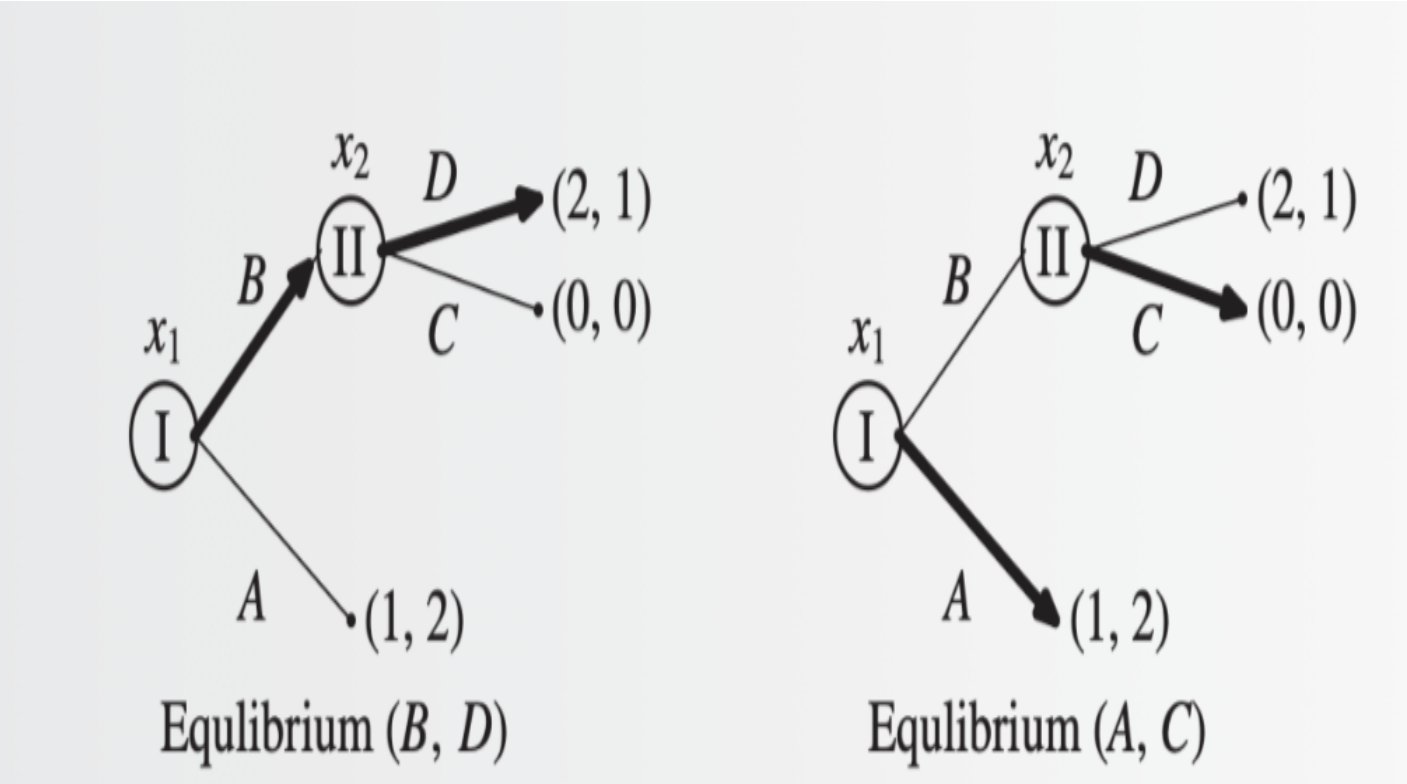

Figure: Figure 1

Example

- Note that this game has two equilibrium (B, D) and (A, C).
- Out of the two, (B, D) is a subgame perfect equilibrium but (A, C) is not.
- This game has two subgames, one starting at x_1 and the other starting at x_2.
- The equilibrium path for (A, C) is $x_1 \to (1, 2)$, whereas for (B, D) is $x_1 \to x_2 \to (2, 1)$.

Example

- Observe that the only non-trivial subgame starting at x_2, and x_2 lies on the equilibrium path of (B, D) not on the equilibrium path of (A, C).
- This is a general fact, for any pure strategy equilibrium s, if a node x lies on the equilibrium path of s, then s restricted to the subgame (if any) starting at x is an equilibrium of the subgame.

Question

- What happens when a mixed (or behavior) equilibrium is played? Or when there are chance nodes in the game?
- Let's see an example.

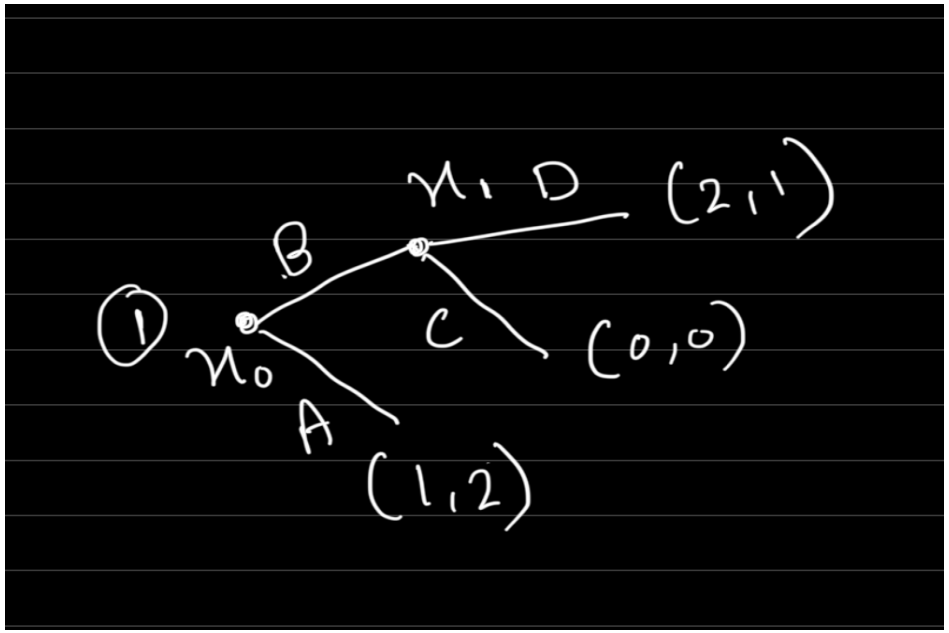

Figure: Figure 2

Example

- We can find the NE in mixed strategies by writing the game in strategic form.

Table: Strategic form game

	Player 2	
Player 1	C	D
A	(1,2)	(1,2)
B	(0,0)	(2,1)

Example

- The set of all NE includes $(A, (\alpha[C], (1-\alpha)[D]))$ where $\alpha \geq \frac{1}{2}$ and $(B, D))$.
- However, any equilibrium of the form $(A, (\alpha[C], (1-\alpha)[D]))$ with $\alpha \geq \frac{1}{2}$ is not a subgame perfect equilibrium as it chooses C with positive probability in the subgame starting at x_1.
- Also, observe that the node x_1 is not reached in the game when $(A, (\alpha[C], (1-\alpha)[D]))$ with $\alpha \geq \frac{1}{2}$ is played. In other words, x_1 is reached with probability 0.

Example

- Let's consider another game. Here, the only change is that the utility after D is now $(1, 1)$.

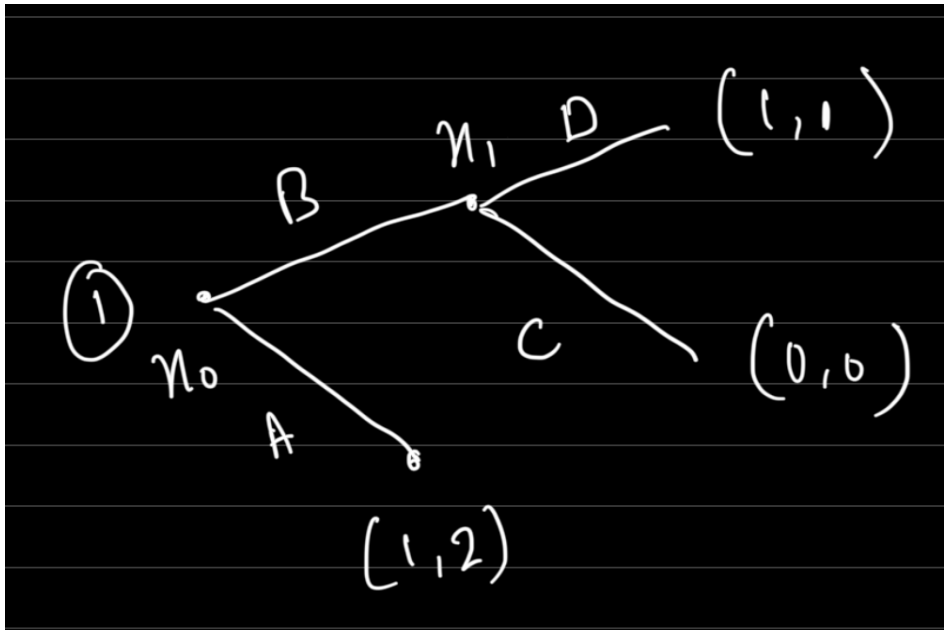

Figure: Figure 3

Example

- Consider the following equilibria: (A, C) and $((\frac{1}{2}[A], \frac{1}{2}[B]), D)$.
- Note that out of these two, the first one (A, C) does not constitute an NE in the subgame starting at x_1. Moreover, under this strategy profile, the probability that the subgame starting at x_1 will be reached is 0.
- The other equilibrium, $((\frac{1}{2}[A], \frac{1}{2}[B]), D)$ is an NE in the subgame starting at x_1, as Player 2 plays D there which he prefers over C. Further, under this equilibrium, the subgame starting at x_1 is reached with probability $\frac{1}{2}$.

Example

- The previous observation holds in general as well.
- For each strategy vector σ, and each node x in the game tree, denote by $P_\sigma(x)$ the probability that the play of the game will visit node x when the players implement the strategy vector σ.
- Recall that we learnt to calculate these probabilities while discussing the equivalence of mixed and behavior strategies.

Result

Theorem

Let $\sigma^\star$ be a Nash equilibrium (in mixed strategies or behavior strategies) of an extensive-form game Γ, and let $\Gamma(x)$ be a subgame of Γ. If $P_{\sigma^\star}(x) > 0$, then the strategy vector $\sigma^\star$ restricted to the subgame $\Gamma(x)$ is a Nash equilibrium (in mixed strategies or behavior strategies) of $\Gamma(x)$.

- The proof can be found in Solan's book, page 255.

Idea of the proof

- Assume for contradiction, σ_* restricted to the subgame $\Gamma(x)$ is not an NE of $\Gamma(x)$. This means there exists a player who can benefit from deviating to some other strategy in the subgame. Let's assume the player be $PPlayer_i$, and the strategy be σ'_i.
- As the subgame reaches with positive probability, Player i can do better in the original game Γ by sticking to $\sigma^\star_i$ everywhere else, and playing σ'_i if the game reaches Γ.

Discussion

- Note that the previous result can sometimes be useful to find subgame perfect equilibria.
- If σ is an NE, and all the subgames in the game are reached with positive probability when σ is implemented. Then σ is also a subgame perfect equilibrium.
- In general, for an NE, the subgames that are reached with zero probability need to be checked separately to conclude if the strategy profile is a subgame perfect equilibrium.

Backward induction principle

- Recall that for perfect information games, we can use the backward induction principle to find all subgame perfect equilibria.
- The same idea can be extended here as well.

Example

- Consider the following example.

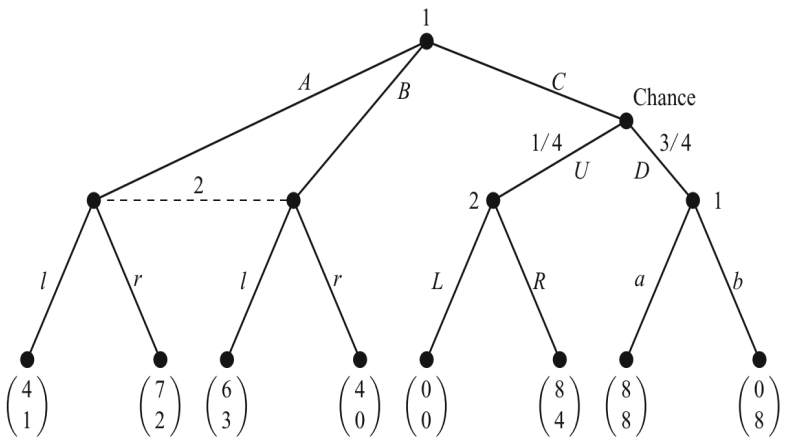

Figure: Figure 4

- Note that there are four subgames here: the entire game; the two subgames following U and D; and the subgame starting from the chance move.

Example

- In this case, backward induction cannot be applied to the left part of the game tree: since Player 2 does not know whether Player 1 has played A or B when he has to choose an action in his left information set, he cannot optimally choose between l and r: l is better if Player 1 has played B, but r is better if Player 1 has played A.
- The restrictions imposed by subgame perfection are that Player 1 should play a, Player 2 should play R, and the strategy combination should be a Nash equilibrium in the entire game.

Example

- Thus, we get the following reduced game.

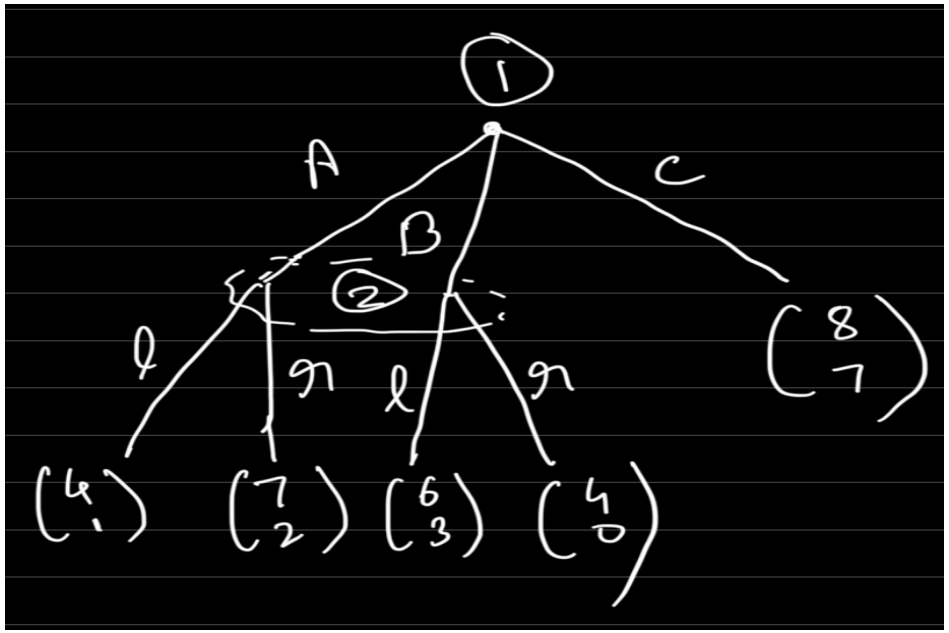

Figure: Figure 5

Example

- Note that the reduced game has only one subgame now, the entire game.
- Therefore, we find the NE of this game to find subgame perfect equilibria of the original game.
- It can be checked that there are two subgame perfect equilibria in pure strategies (Ca, lR) and (Ca, rR).
- Similarly, we can find the subgame perfect equilibria in mixed strategies by computing the mixed NE of the reduced game.

Result

- The above procedure can be generalized for all imperfect information games. In particular, we have the following theorem.

Theorem

Every extensive-form game has a subgame perfect equilibrium in mixed strategies.

Limitations of subgame perfect equilibria

- Recall that an NE may not always ensure playing rationally in all nodes of the game.
- That's the motivation behind introducing the idea of subgame perfect equilibrium.
- A subgame perfect equilibrium serves the purpose for perfect information games.
- However, it may not ensure rationality in all information sets of an imperfect information game.

Limitations of subgame perfect equilibria

- Consider the following game.

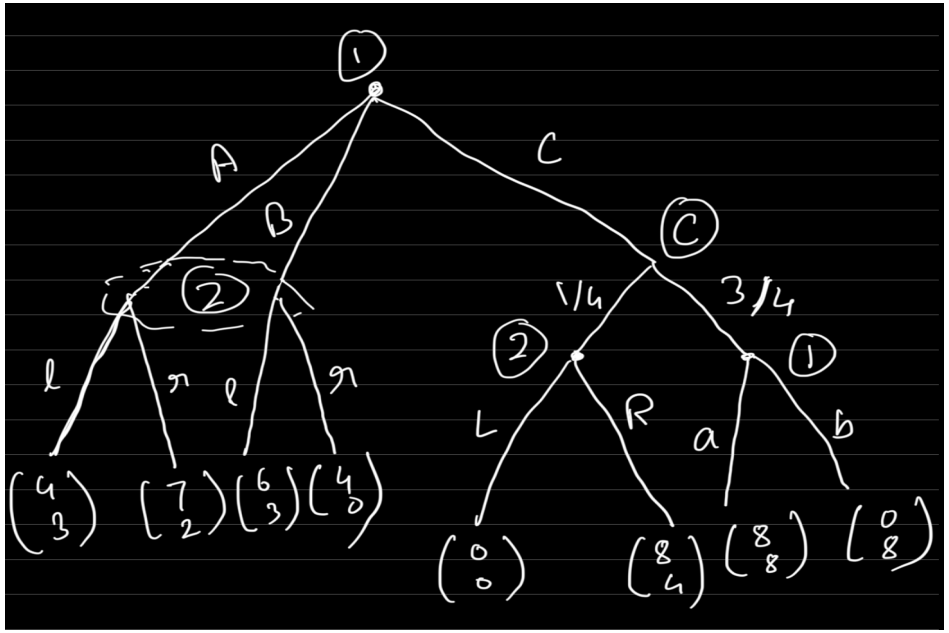

Figure: Figure 6

- One may check that the game has a subgame perfect equilibrium (Ca, rR).

Limitations of subgame perfect equilibria

- However, observe that a rational player (Player 2) will never play r at the only non-trivial information set in the game, as l is always better.
- But subgame perfection does not rule this out.
- This observation leads to further refinements of the notion of subgame perfect equilibria, also known as "perfect Bayesian equilibrium". But we will not discuss this in this course.

Further discussion

- Kohlberg and Mertens [1986] argued that backward induction requires that at each stage, every player looks only at the continuation of the game from that stage forwards, and ignores the fact that the game has reached that stage. But if the game has reached a particular vertex in the game tree, that fact itself gives information about the behavior of the other players, and this should be taken into account.
- In the next example, we will see how this may change the way players may play even in perfect information games.

Example

- Consider the following game.

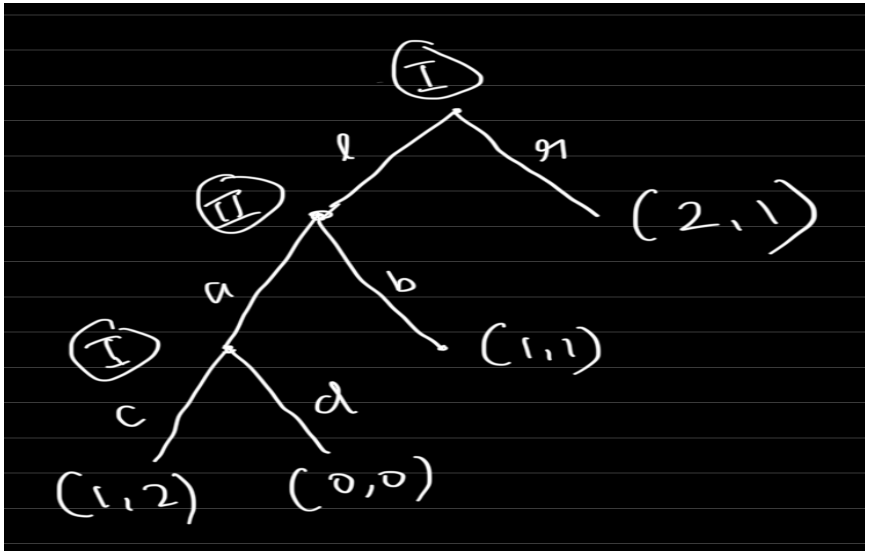

Figure: Figure 7

- The only subgame perfect equilibrium of this game is (rc, a), which yields the payoff $(2, 1)$.

Example

- Why does Player II choose a at x_2? Because of the assumption that Player I is rational, he will then choose c, leading to the payoff $(1, 2)$, which Player II prefers to the payoff $(1, 1)$ that would result if he were to choose b. But is Player I really rational?
- Consider the fact that if the game has arrived at x_2, and Player II is called upon to play, then Player I must be irrational: Player I must have chosen l, which yields him at most 1, instead of choosing r, which guarantees him 2. Then why should Player II assume that Player I will be rational at x_3?

Example

- Perhaps it would be more rational for Player II to choose b, and guarantee himself a payoff of 1, instead of running the risk that Player I may again be irrational and choose d, which will yield the payoff $(0, 0)$.

Example

- The situation would be even more problematic if we replace the utility $(1, 1)$ by $(3, 1)$.
- A similar argument to the one before may lead Player II to play b if he is called upon to play at his information set.
- However, then Player I is better off by playing l than playing r (his rational choice), as playing r will give him 2, but playing l, the outcome is $(3, 1)$.

Example

- So is Player I really irrational if he chooses l? Perhaps Player I's choice of l is a calculated choice, aimed at making Player II think that he is irrational, and therefore leading Player II to choose b?
- Then, which one of Player I's choices at vertex x_1 is rational, and which is irrational?

The relationship between the extensive form and the strategic form

- Note that we have already seen that from an extensive form game, we may derive a strategic form game uniquely.
- Now, what about the opposite direction?
- Does every strategic-form game have an extensive-form game from which it is derived? And, is there a unique extensive-form game associated with a strategic-form game?

The relationship between the extensive form and the strategic form

- The answer to the first question is yes, whereas the second question has a negative answer.
- Suppose $G = (N, (S_i))_{i \in N}, (u_i)_{i \in N})$ is strategic form game with strategy set $S_i = \{s_i^1, \ldots, s_i^{m_i}\}$ for each i.

Writing a strategic form in an extensive form game

- Consider the following extensive form game:

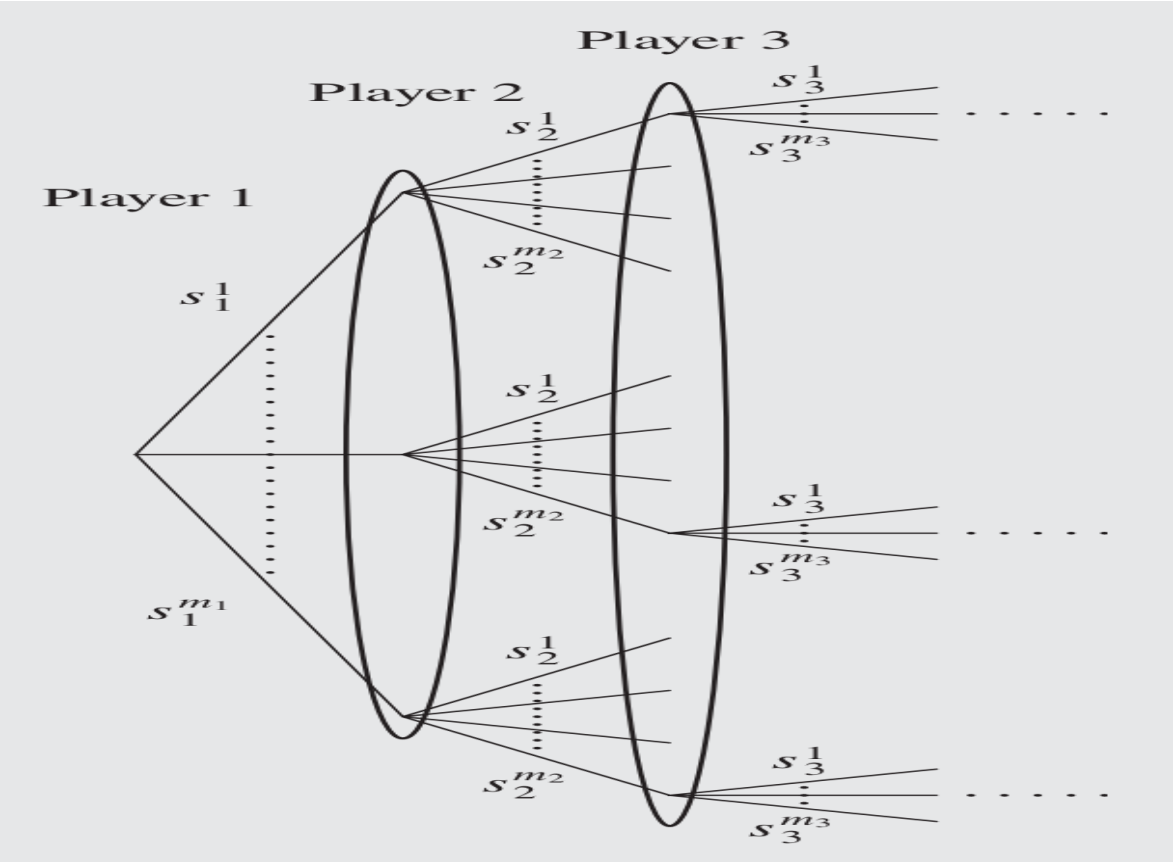

Figure: Figure 8

Writing a strategic form in an extensive form game

- It can be verified that the associated strategic form game of the extensive form game in Figure 8 is G.
- Further, the representation is not unique, as the ordering of the agents can be changed, and there are $n!$ such games possible.

Example

- Consider the following game:

Table: Strategic form game

Player 1	Player 2			
	rt	rb	lt	lb
L	(2,-1)	(2,-1)	(2,0)	(2,0)
R	(3,1)	(1,0)	(3,1)	(1,0)

Example

- Note that other than having a representation as shown in Figure 8, the following extensive form also represents the above strategic form game.

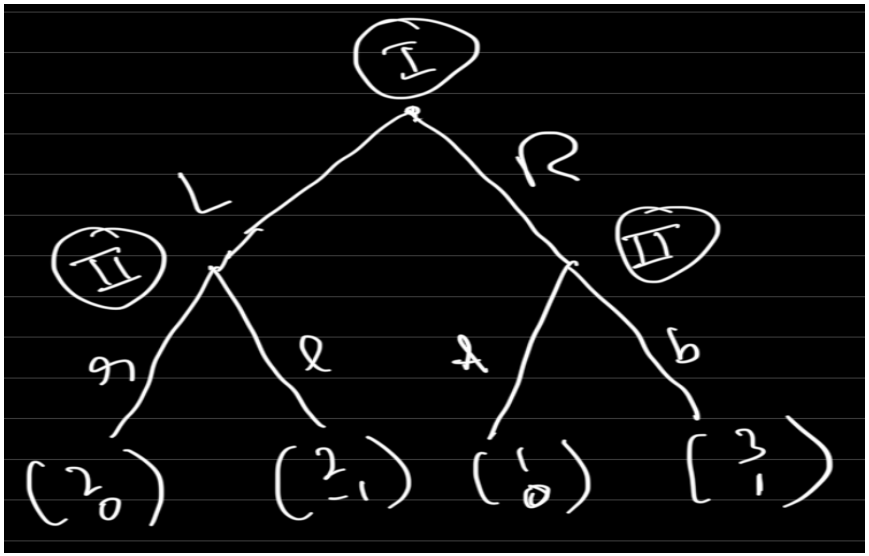

Figure: Figure 9

Discussion

- In general, one may ask the question, is there a description of all the extensive form games that represent the same strategic form game?
- This subject was studied by Thompson [1952], who defined three elementary operations that do not change the "essence" of a game.
- He then proved that if two games in extensive form with the same set of players can be transformed into each other by a finite number of these three elementary operations, then those two extensive-form games correspond to the same strategic-form game.

Discussion

- He also showed that the other direction obtains: if two games in extensive form yield the same In a strategic-form game, then, they can be transformed into each other by a finite number of these three elementary operations.